Purchase Orders Project

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Date: 29th April 2025

DOCUMENTATION

1 OVERVIEW:

By now you are familiar with most of the PowerCenter functionalities and can sufficiently utilize them to develop a project that suits the business needs. In retail for example, for performance analysis, the business will use the raw data they possess to generate meaningful performance checks for its employees and business as a whole in order better understand and maintain the sustainability of the business, and exploit any weaknesses.

2 SCENARIO:

You are asked to create 3 target tables **ODS_PO**, **EMP_ORDERS**, **CUST_ORDERS** according to below logic:

- **ODS_PO**: load all purchase orders adding all info for customers and employees
- **EMP_ORDERS**: load orders aggregated per employee, employee id and name should be available in the table
- **CUST_ORDERS**: load orders aggregated per customer, customer id and name should be available in the table

Points to consider:

- **ODS_PO** should be loaded first, if succeeded, load the other 2, otherwise, load process should fail.

3 GOAL:

Get familiar with creating a complete project from scratch using the Informatica PowerCenter, utilizing the transformations in the Designer and the tasks in the Workflow Manager learned through the first training guide (PWC Training L1).

4 RESOURCES:

Source Table(s): CUST, EMP

Source File(s): po_data.csv

Source System Schema: SRC_SYS @ ORCLDB

Login: SRC_SYS/ SRC_SYS

Project methodology:

The project will consist of a set of parts as following:

1. Database setup
2. Mapping design
3. Workflow creation
4. Output

1. Database setup:

The first part of the project is to setup our database as it will be consisting from 2 part as source and target tables

Source setup:

the first step is to setup the source as we have 3 data sources, first one is po_data.csv which contain purchase order data, second one is EMP data which contain information about employees and last one is CUST data which contain information about customers, the following snapshots will show the three different data sources:

PO_SERIAL	PO_BILL_NUMBER	PO_DATE	PO_EMP_ID	PO_CUST_ID	PO_ITEM_DESC	PO_QUANTITY	PO_UNIT_PRICE	PO_TOTAL_PRICE
100001	100001-782767-001	20080111	782767	1	Bill: 100001-782767-001, Item: 1	1	10	10
100002	100002-782767-001	20080111	782767	1	Bill: 100002-782767-001, Item: 2	1	12	12
100003	100003-978283-003	20010101	978283	3	Bill: 100003-978283-003, Item: 1	1	145	145
100004	100004-100220-004	20180612	100220	4	Bill: 100004-100220-004, Item: 1	1	23	23
100005	100005-102345-005	20160523	102345	5	Bill: 100005-102345-005, Item: 1	1	45	45
100006	100006-102345-006	20110109	102345	6	Bill: 100006-102345-006, Item: 1	1	65	65
100007	100007-100220-007	20000102	100220	7	Bill: 100007-100220-007, Item: 1	1	89	89
100008	100008-100220-008	20180823	100220	8	Bill: 100008-100220-008, Item: 1	1	800	800
100009	100009-100220-009	20170723	100220	9	Bill: 100009-100220-009, Item: 1	1	10000	10000
100010	100010-789578-010	20120101	789578	10	Bill: 100010-789578-010, Item: 1	1	12	12
100011	100011-123238-002	20120304	123238	2	Bill: 100011-123238-002, Item: 1	1	43	43
100012	100012-782767-008	20130908	782767	8	Bill: 100012-782767-008, Item: 1	1	24	24
100013	100013-102345-009	20101002	102345	9	Bill: 100013-102345-009, Item: 1	1	90	90
100014	100014-203748-010	20080304	203748	10	Bill: 100014-203748-010, Item: 1	1	60	60
100015	100015-203748-007	20150423	203748	7	Bill: 100015-203748-007, Item: 3	1	40	40
100016	100016-203748-007	20150423	203748	7	Bill: 100016-203748-007, Item: 1	1	1234	1234
100017	100017-203748-007	20150423	203748	7	Bill: 100017-203748-007, Item: 2	1	435	435
100018	100018-123238-002	20000505	123238	2	Bill: 100018-123238-002, Item: 1	1	60	60
100019	100019-123238-003	20140626	123238	3	Bill: 100019-123238-003, Item: 1	1	80	80
100020	100020-102345-004	20180101	102345	4	Bill: 100020-102345-004, Item: 1	1	90	90
100021	100021-100220-001	20180109	100220	1	Bill: 100021-100220-001, Item: 1	1	100	100
100022	100022-782767-003	20181002	782767	3	Bill: 100022-782767-003, Item: 1	1	120	120
100023	100023-789578-004	20180909	789578	4	Bill: 100023-789578-004, Item: 1	1	320	320
100024	100024-324315-005	20180304	324315	5	Bill: 100024-324315-005, Item: 1	1	560	560
100025	100025-789578-006	20170403	789578	6	Bill: 100025-789578-006, Item: 1	1	700	700

Figure 1- po_data

EMP_ID	FULL_NAME	FULL_NAME_OL	HIRE_DATE	EMAIL	TELEPHONE_NUMBER
100220	Emp 1	موظف 1	1/1/2001	emp.1@company.com	
102345	Emp 2	موظف 2	1/1/2001	emp.2@company.com	
203748	Emp 3	موظف 3	1/1/2001	emp.3@company.com	
324315	Emp 4	موظف 4	1/1/2001	emp.4@company.com	
782767	Emp 5	موظف 5	1/1/2001	emp.5@company.com	
789578	Emp 6	موظف 6	1/1/2001	emp.6@company.com	
978283	Emp 7	موظف 7	1/1/2001	emp.7@company.com	
123238	Emp 8	موظف 8	1/1/2001	emp.8@company.com	

Figure 2- EMP Table

CUST_ID	FULL_NAME	FULL_NAME_OL	CUST_SINCE_DATE	ADDRESS	CUST_TYPE	EMAIL	TELEPHONE_NUMBER
001	Farid Eldakrory	فريد الدكروري	01/02/2017	xyz	R	abcd@xyz.xyz	xxxxxxxxxxx
002	Mohammed Fawzy	محمد فوزي	20070225	8 Galeel Elbendari St. - Sporting, Alexandria	R	abcd@xyz.xyz	00000000000
003	Ali Abdullah	علي عبدالله	27/02/2018		C	abcd@xyz.xyz	
004	XYZ Company	شركة إكس واي زد	01/01/2000	سموحة	C		
005	Gad Restaurant	مطاعم جاد	24-Mar-2005	Stanley, elkornesh	C	abcd@xyz.xyz	
006	Heba Ahmed	هبة أحمد	21/05/2018		R	abcd@xyz.xyz	xxxxxxxxxxx
007	Ahmed Said	أحمد سعيد	01/01/2011	Sidi Bishr	R		
008	Sayed Elshamandori	سيد الشمندوري	20030426	Maadi, Cairo	C		
009	ABC For Import & Export	إيه بي سي للإستيراد و التصدير	22/02/2010	Moharram Bek, Alexandria	C	abcd@xyz.xyz	xxxxxxxxxxx
010	Khaled Yehia	خالد يحيي	28/06/2015	سبورتنج - الإسكندرية	R	abcd@xyz.xyz	11111111111

Figure 3 - CUST Table

After we saw the data sources we have, it’s time to create a database table for each one, one for EMP data and another one for CUST data and po_data.csv will be loaded directly later.

so we create two tables called EMP and CUST to store both employee and customer’s data using SQL query stored in Source Query.sql file as following:

```sql
-- Create employee table
CREATE TABLE EMP (
   EMP_ID          INTEGER,
   FULL_NAME       VARCHAR2(100 CHAR),
   FULL_NAME_OL    VARCHAR2(100 CHAR),
   HIRE_DATE DATE,
   EMAIL       VARCHAR2(50 char),
   TELEPHONE_NUMBER VARCHAR2(50 char)
);

-- Insert data into employee table
insert into EMP values (100220, 'Emp 1', 'موظف 1', TO_DATE('1/1/2001', 'MM/DD/YYYY'), 'emp.1@company.com', '');
insert into EMP values (102345, 'Emp 2', 'موظف 2', TO_DATE('1/1/2001', 'MM/DD/YYYY'), 'emp.2@company.com', '');
insert into EMP values (203748, 'Emp 3', 'موظف 3', TO_DATE('1/1/2001', 'MM/DD/YYYY'), 'emp.3@company.com', '');
insert into EMP values (324315, 'Emp 4', 'موظف 4', TO_DATE('1/1/2001', 'MM/DD/YYYY'), 'emp.4@company.com', '');
insert into EMP values (782767, 'Emp 5', 'موظف 5', TO_DATE('1/1/2001', 'MM/DD/YYYY'), 'emp.5@company.com', '');
insert into EMP values (789578, 'Emp 6', 'موظف 6', TO_DATE('1/1/2001', 'MM/DD/YYYY'), 'emp.6@company.com', '');
insert into EMP values (978283, 'Emp 7', 'موظف 7', TO_DATE('1/1/2001', 'MM/DD/YYYY'), 'emp.7@company.com', '');
insert into EMP values (123238, 'Emp 8', 'موظف 8', TO_DATE('1/1/2001', 'MM/DD/YYYY'), 'emp.8@company.com', '');
```

Figure 4 - EMP table creation

```sql
-- Create customer table
CREATE TABLE CUST (
   CUST_ID          INTEGER,
   FULL_NAME        VARCHAR2(100 CHAR),
   FULL_NAME_OL     VARCHAR2(100 CHAR),
   CUST_SINCE_DATE VARCHAR2(20 char),
   ADDRESS      VARCHAR2(200 CHAR),
   CUST_TYPE        VARCHAR2(1 CHAR),
   EMAIL       VARCHAR2(50 char),
   TELEPHONE_NUMBER VARCHAR2(50 char)
);

-- Insert data into customer table
insert into CUST values (001, 'Farid Eldakrory', 'فريد الدكروري', '01/02/2017', 'xyz', 'R', 'abcd@xyz.xyz', 'xxxxxx
insert into CUST values (002, 'Mohammed Fawzy', 'محمد فوزي', '20070225', '8 Galeel Elbendari St. - Sporting, Alexan
insert into CUST values (003, 'Ali Abdullah', 'علي عبد الله', '27/02/2018', '', 'C', 'abcd@xyz.xyz', '');
insert into CUST values (004, 'XYZ Company', 'شركة إكس واي زد', '01/01/2000', 'سموحة', 'C', '', '');
insert into CUST values (005, 'Gad Restaurant', 'مطاعم جاد', '24-Mar-2005', 'Stanley, elkornesh', 'C', 'abcd@xyz.xy
insert into CUST values (006, 'Heba Ahmed', 'هبة أحمد', '21/05/2018', '', 'R', 'abcd@xyz.xyz', 'xxxxxxxxxxx');
insert into CUST values (007, 'Ahmed Said', 'أحمد سعيد', '01/01/2011', 'Sidi Bishr', 'R', '', '');
insert into CUST values (008, 'Sayed Elshamandori', 'سيد الشمندوري', '20030426', 'Maadi, Cairo', 'C', '', '');
insert into CUST values (009, 'ABC For Import and Export', 'إيه بي سي للإستيراد و التصدير' , '22/02/2010', 'Moharram
insert into CUST values (010, 'Khaled Yehia', 'خالد يحيى', '28/06/2015', 'سبورتنج - الإسكندرية' , 'R', 'abcd@xyz.xyz
```

Figure 5 - CUST table creation

Source setup:

the second step is to setup the target as we have 3 target:

- first one is ODS_PO which contain purchase order data along with employees and customer's data,

- second one is EMP_ORDERS which contain information about employees aggregated by their orders.
- last one is CUST_ORDERS which contain information about customers aggregated by their orders.

so we create three tables called ODS_PO, EMP_ORDERS and CUST_ORDERS using SQL query stored in Target_Query.sql file as following:

```sql
-- Create ODS_PO table
CREATE TABLE ODS_PO (
	PO_SERIAL			NUMBER,
	PO_BILL_NUMBER			VARCHAR2(26),
	PO_DATE				NUMBER,
	PO_EMP_ID			NUMBER,
	PO_CUST_ID			NUMBER,
	PO_ITEM_DESC			VARCHAR2(128),
	PO_QUANTITY			NUMBER,
	PO_UNIT_PRICE			NUMBER,
	PO_TOTAL_PRICE			NUMBER,
	EMP_EMP_ID			VARCHAR2(20),
	EMP_FULL_NAME			VARCHAR2(20),
	EMP_FULL_NAME_OL		VARCHAR2(20),
	EMP_HIR_DATE			DATE,
	EMP_EMAIL			VARCHAR2(20),
	EMP_TELEPHONE_NUMBER		VARCHAR2(20),
	CUST_CUST_ID			VARCHAR2(20),
	CUST_FULL_NAME			VARCHAR2(80),
	CUST_FULL_NAME_OL		VARCHAR2(80),
	CUST_CUST_SINCE_DATE		VARCHAR2(20),
	CUST_ADDRESS			VARCHAR2(150),
	CUST_CUST_TYPE			VARCHAR2(20),
	CUST_EMAIL			VARCHAR2(20),
	CUST_TELEPHONE_NUMBER		VARCHAR2(20)
);
```

Figure 6 - ODS_PO table creation

```sql
-- Create EMP_ORDERS table
CREATE TABLE EMP_ORDERS (
	EMP_ID			NUMBER,
	EMP_NAME		VARCHAR2(100),
	EMP_NAME_OL		VARCHAR2(100),
	TOTAL_ORDERS		INTEGER
);
```

Figure 7 - EMP_ORDERS table creation

```sql
-- Create CUST_ORDERS table
CREATE TABLE CUST_ORDERS (
	CUST_ID			NUMBER,
	CUST_NAME		VARCHAR2(100),
	CUST_NAME_OL		VARCHAR2(100),
	TOTAL_ORDERS		INTEGER
);
```

Figure 8 - CUST_ORDERS table creation

2. Mapping design:

The second part of the project is to design mappings using informatica powercenter designer to design an ETL maps for the following tasks:

2.1. PO data loading: this task is to load the three sources data and join them together and load them in a target table called ODS_PO, the following snapshot show the mapping called M_ PO_DATA_LOADING which designed for this task:

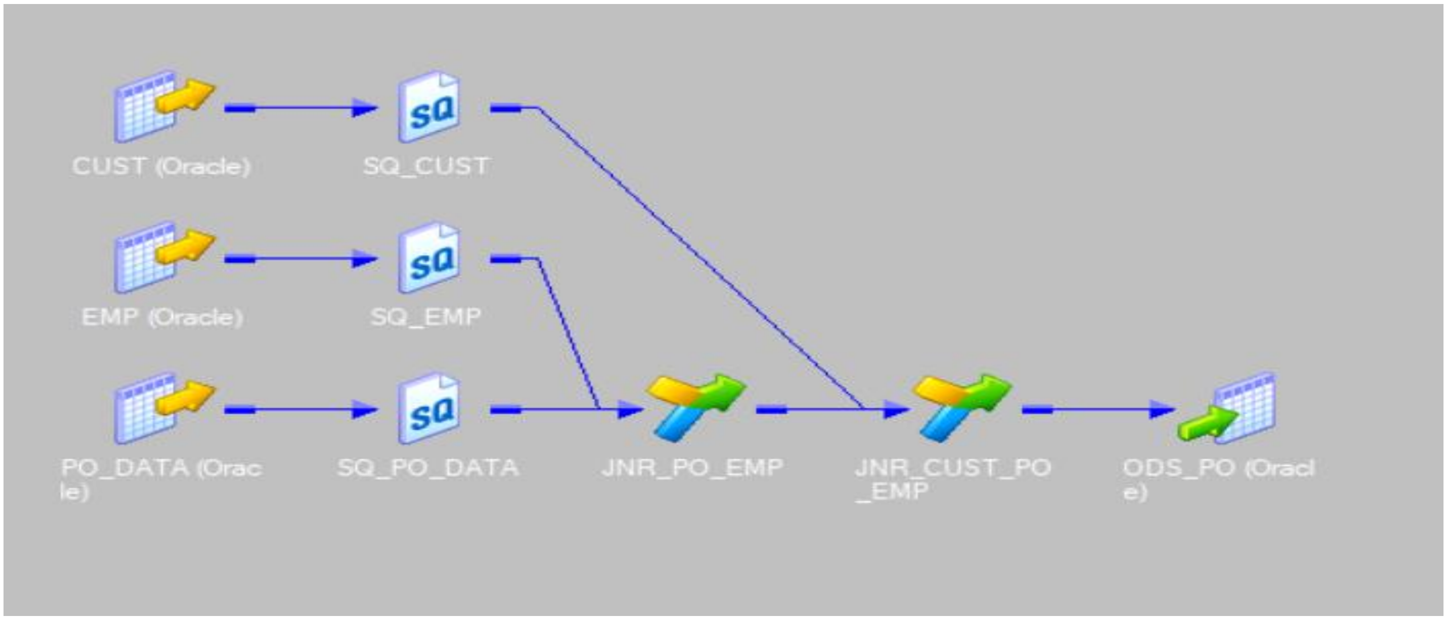

Figure 9 - M_ PO_DATA_LOADING mapping

2.2. Employee orders: this task is to use the target we loaded ODS_PO to aggregate orders per employee, the following snapshot show the mapping called M_EMP_ORDERS which designed for this task:

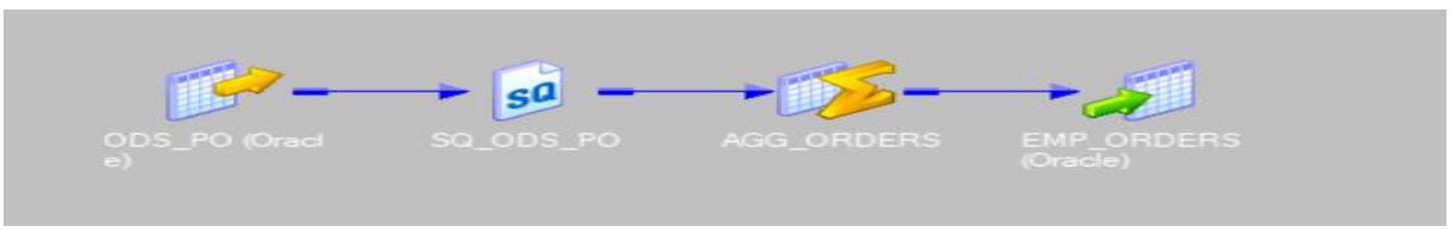

Figure 10 - M_EMP_ORDERS mapping

2.3. Customers orders: this task is to use the target we loaded ODS_PO to aggregate orders per customer, the following snapshot show the mapping called M_CUST_ORDERS which designed for this task:

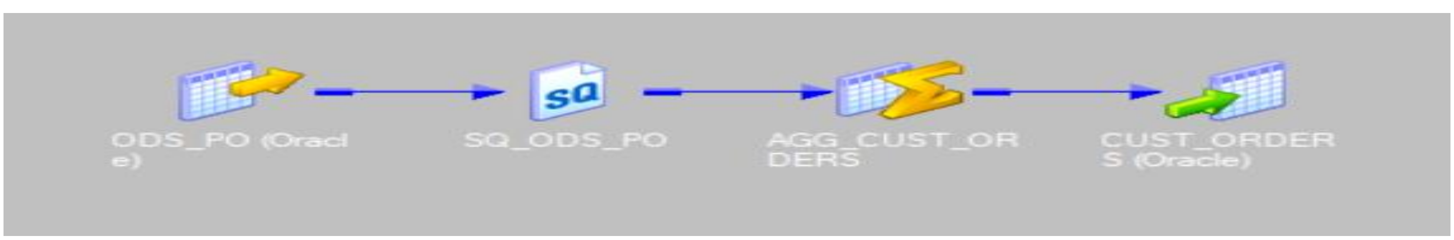

Figure 11 - M_CUST_ORDERS mapping

3. Workflow creation:

The third part of the project is to create a set of workflows which will use mappings we created to execute these mappings and generate outputs.

We create three mappings one for each mapping as following:

- WF_ODS_PO
- WF_EMP_ORDERS
- WF_CUST_ORDERS

The following snapshot shows the three workflows:

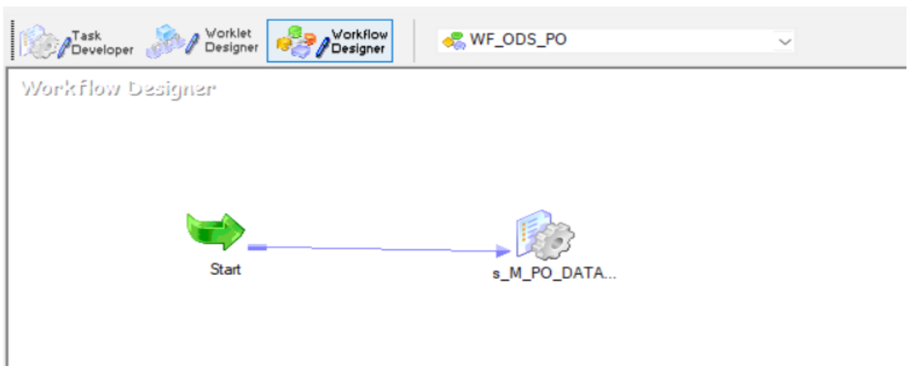

Figure 12 - WF_ODS_PO workflow

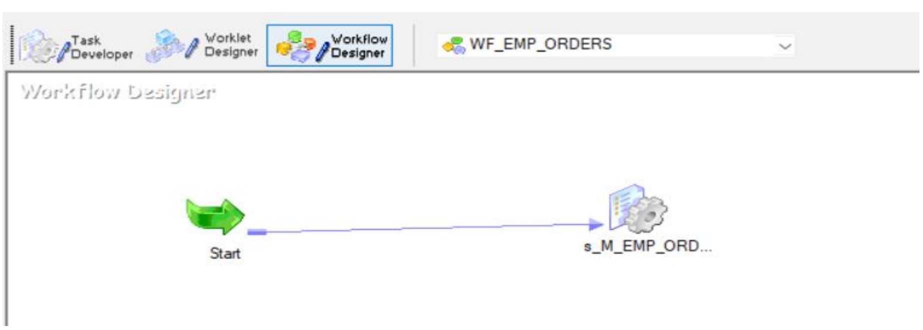

Figure 13 - WF_EMP_ORDERS workflow

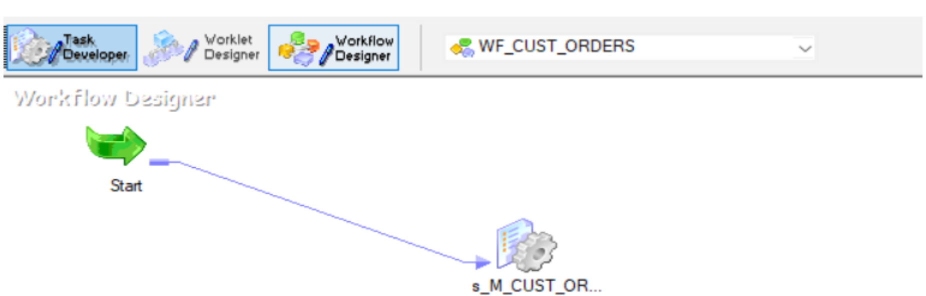

Figure 14 - WF_CUST_ORDERS workflow

4. Output:

The last part of the project is to show the output of the workflows we created, as we have three targets ODS_PO, EMP_ORDERS and CUST_ORDERS, the following snapshots will show the three tables after executing all workflows:

	PO_SE...	PO_BILL_NUMBER	PO_DATE	PO_EMP_ID	PO_CUST_ID	PO_ITEM_DESC	PO_QUANTITY	PO_UNIT_PRICE	PO_TOTAL_PRICE	EMP_EMP_ID	EMP_FULL_NAME	EMP_FULL_NAME_OL	EMP_HIR_DATE	EM
1	100001	100001-782767-001	20080111	782767	1	Bill: 100001-782767-001, Item: 1	1	10	10	782767.000000000000	Emp 5	5 موظف	01-JAN-01	em
2	100002	100002-782767-001	20080111	782767	1	Bill: 100002-782767-001, Item: 2	1	12	12	782767.000000000000	Emp 5	5 موظف	01-JAN-01	em
3	100003	100003-978283-003	20010101	978283	3	Bill: 100003-978283-003, Item: 1	1	145	145	978283.000000000000	Emp 7	7 موظف	01-JAN-01	em
4	100004	100004-100220-004	20180612	100220	4	Bill: 100004-100220-004, Item: 1	1	23	23	100220.000000000000	Emp 1	1 موظف	01-JAN-01	em
5	100005	100005-102345-005	20160523	102345	5	Bill: 100005-102345-005, Item: 1	1	45	45	102345.000000000000	Emp 2	2 موظف	01-JAN-01	em
6	100006	100006-102345-006	20110109	102345	6	Bill: 100006-102345-006, Item: 1	1	65	65	102345.000000000000	Emp 2	2 موظف	01-JAN-01	em
7	100007	100007-100220-007	20000102	100220	7	Bill: 100007-100220-007, Item: 1	1	89	89	100220.000000000000	Emp 1	1 موظف	01-JAN-01	em
8	100008	100008-100220-008	20180823	100220	8	Bill: 100008-100220-008, Item: 1	1	800	800	100220.000000000000	Emp 1	1 موظف	01-JAN-01	em
9	100009	100009-100220-009	20170723	100220	9	Bill: 100009-100220-009, Item: 1	1	10000	10000	100220.000000000000	Emp 1	1 موظف	01-JAN-01	em
10	100010	100010-789578-010	20120101	789578	10	Bill: 100010-789578-010, Item: 1	1	12	12	789578.000000000000	Emp 6	6 موظف	01-JAN-01	em
11	100011	100011-123238-002	20120304	123238	2	Bill: 100011-123238-002, Item: 1	1	43	43	123238.000000000000	Emp 8	8 موظف	01-JAN-01	em
12	100012	100012-782767-008	20130908	782767	8	Bill: 100012-782767-008, Item: 1	1	24	24	782767.000000000000	Emp 5	5 موظف	01-JAN-01	em
13	100013	100013-102345-009	20101002	102345	9	Bill: 100013-102345-009, Item: 1	1	90	90	102345.000000000000	Emp 2	2 موظف	01-JAN-01	em
14	100014	100014-203748-010	20080304	203748	10	Bill: 100014-203748-010, Item: 1	1	60	60	203748.000000000000	Emp 3	3 موظف	01-JAN-01	em
15	100015	100015-203748-007	20150423	203748	7	Bill: 100015-203748-007, Item: 3	1	40	40	203748.000000000000	Emp 3	3 موظف	01-JAN-01	em
16	100016	100016-203748-007	20150423	203748	7	Bill: 100016-203748-007, Item: 1	1	1234	1234	203748.000000000000	Emp 3	3 موظف	01-JAN-01	em
17	100017	100017-203748-007	20150423	203748	7	Bill: 100017-203748-007, Item: 2	1	435	435	203748.000000000000	Emp 3	3 موظف	01-JAN-01	em
18	100018	100018-123238-002	20000505	123238	2	Bill: 100018-123238-002, Item: 1	1	60	60	123238.000000000000	Emp 8	8 موظف	01-JAN-01	em
19	100019	100019-123238-003	20140626	123238	3	Bill: 100019-123238-003, Item: 1	1	80	80	123238.000000000000	Emp 8	8 موظف	01-JAN-01	em
20	100020	100020-102345-004	20180101	102345	4	Bill: 100020-102345-004, Item: 1	1	90	90	102345.000000000000	Emp 2	2 موظف	01-JAN-01	em
21	100021	100021-100220-001	20180109	100220	1	Bill: 100021-100220-001, Item: 1	1	100	100	100220.000000000000	Emp 1	1 موظف	01-JAN-01	em
22	100022	100022-782767-003	20181002	782767	3	Bill: 100022-782767-003, Item: 1	1	120	120	782767.000000000000	Emp 5	5 موظف	01-JAN-01	em
23	100023	100023-789578-004	20180909	789578	4	Bill: 100023-789578-004, Item: 1	1	320	320	789578.000000000000	Emp 6	6 موظف	01-JAN-01	em
24	100024	100024-324315-005	20180304	324315	5	Bill: 100024-324315-005, Item: 1	1	560	560	324315.000000000000	Emp 4	4 موظف	01-JAN-01	em
25	100025	100025-789578-006	20170403	789578	6	Bill: 100025-789578-006, Item: 1	1	700	700	789578.000000000000	Emp 6	6 موظف	01-JAN-01	em

Figure 15 - ODS_PO target table (trimmed)

	EMP_ID	EMP_NAME	EMP_NAME_OL	TOTAL_ORDERS
1	100220	Emp 1	1 موظف	5
2	102345	Emp 2	2 موظف	4
3	203748	Emp 3	3 موظف	4
4	324315	Emp 4	4 موظف	1
5	782767	Emp 5	5 موظف	4
6	789578	Emp 6	6 موظف	3
7	978283	Emp 7	7 موظف	1
8	123238	Emp 8	8 موظف	3

Figure 16 - EMP_ORDERS target table

	CUST_ID	CUST_NAME	CUST_NAME_OL	TOTAL_ORDERS
1	1	Farid Eldakrory	فريد الدكرورى	3
2	2	Mohammed Fawzy	محمد فوزى	2
3	3	Ali Abdullah	علي عبد الله	3
4	4	XYZ Company	شركة إكس واى زد	3
5	5	Gad Restaurant	مطعم جاد	2
6	6	Heba Ahmed	هبة أحمد	2
7	7	Ahmed Said	أحمد سعيد	4
8	8	Sayed Elshamandori	سيد الشمندورى	2
9	9	ABC For Import ...	اد و التصدير...	2
10	10	Khaled Yehia	خالد يحيى	2

Figure 17 - CUST_ORDERS target table